

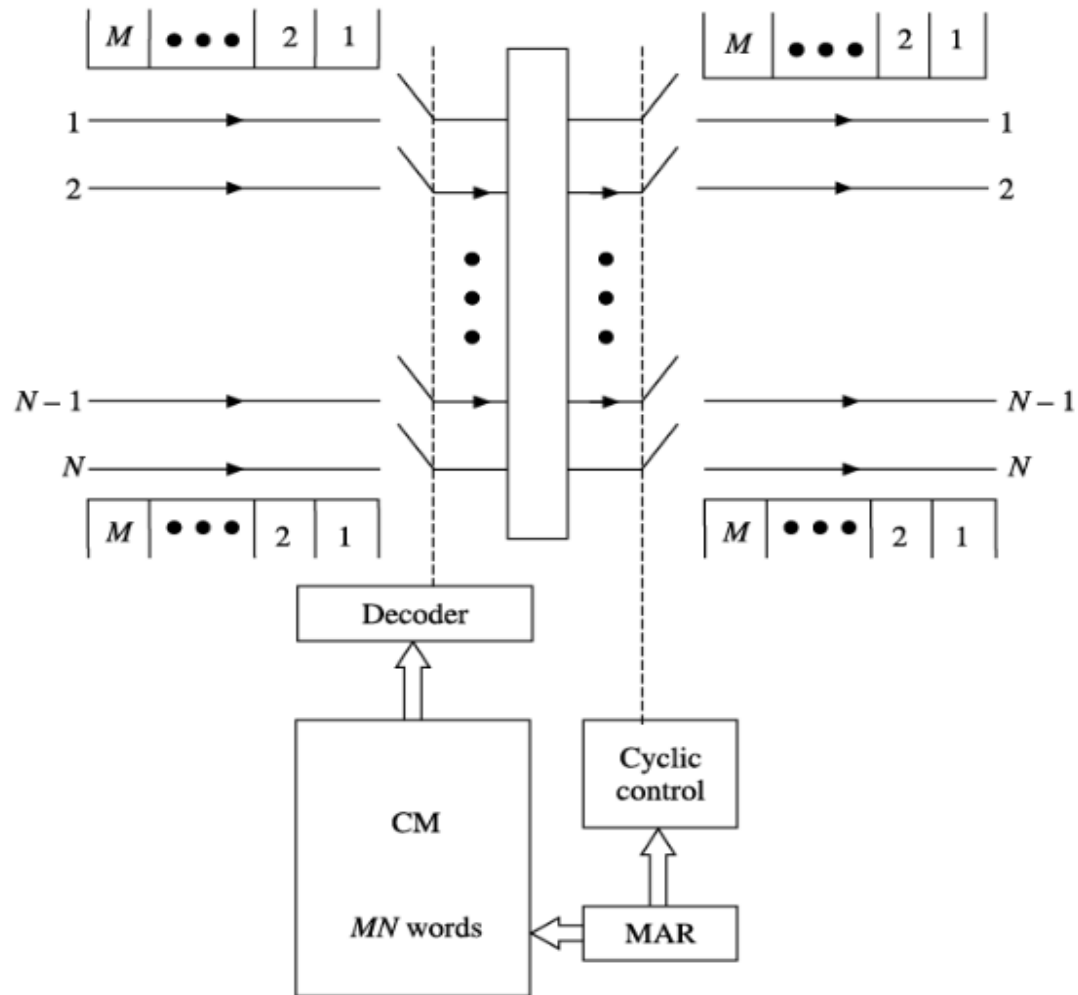
TIME DIVISION SWITCHING

Time Division Switching

❑ Features of PAM/PCM signal

- ✓ Discrete in time domain
- ✓ Transmission of PAM/PCM signal Multiplexing
- ✓ Periodical time slots
- ❑ For PAM/PCM signals, it is possible for a number of active speech circuits to share a single switching element, and therefore reduce the number of switching elements significantly.

Time Multiplexed space switching



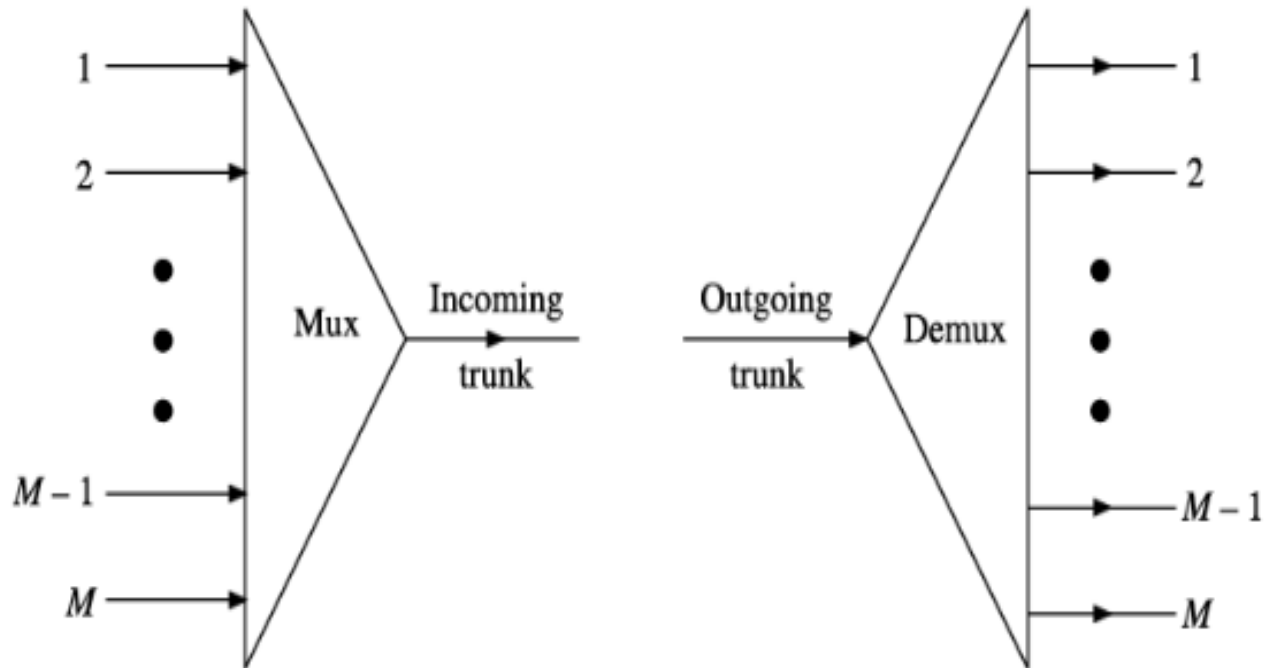
Time Multiplexed space switching

- Time division switches where an inlet or an outlet corresponded to a single subscriber line with one speech sample appearing every 125 μ s on the line.
- Such switches are used in local exchanges.
- The inlets and outlets are trunks which carry time division multiplexed data streams.

Time Division Switching

- They are N incoming trunks and N outgoing trunks, each carrying a time division multiplexed stream of M samples per frame.
- Each frame is of 125- μ s time duration.
- In one frame time, a total of MN speech samples have to be switched.
- One sample duration, $125M$ microseconds, is usually referred to as a time slot

Time Multiplexed space switching



Source multiplexing and destination demultiplexing.

Time Multiplexed space switching

- Sources and incoming trunks time slots.
- Outgoing trunk time slots and the destinations.
- Time slots of incoming and outgoing trunks.

Time Multiplexed Time Switching

- Time multiplexed time switches permit time slot interchange (TSI) of sample values.
- In TSI, a speech sample input during one time slot may be sent to the output during a different time slot.
- Such an operation necessarily implies a delay between the reception and the transmission of a sample.

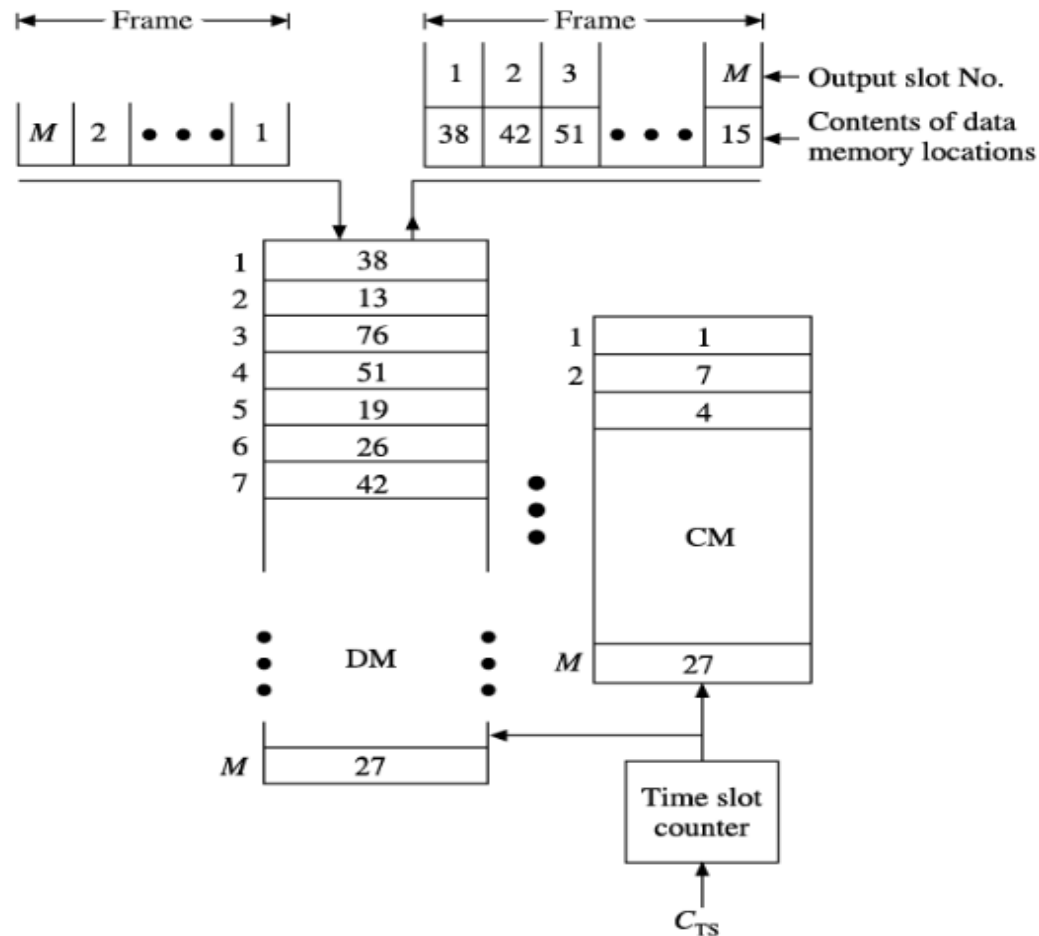
Time Multiplexed Time Switching

- The time slot duration t_{TS} is given by

$$t_{TS} = \frac{125}{M}$$

- The time slot clock runs at the time slot rate i.e at the rate of one pulse ever' 125M microseconds.
- The time slot counter is incremented by one at the end of each time slot.
- The contents of the counter provides location addresses for the data memory and the control memory. Data memory and Control memory accesses take place simultaneously in the beginning of the time slot.
- The contents of the control memory are used as the address of the data memory and the data read out to the output trunk

Time Multiplexed Time Switching



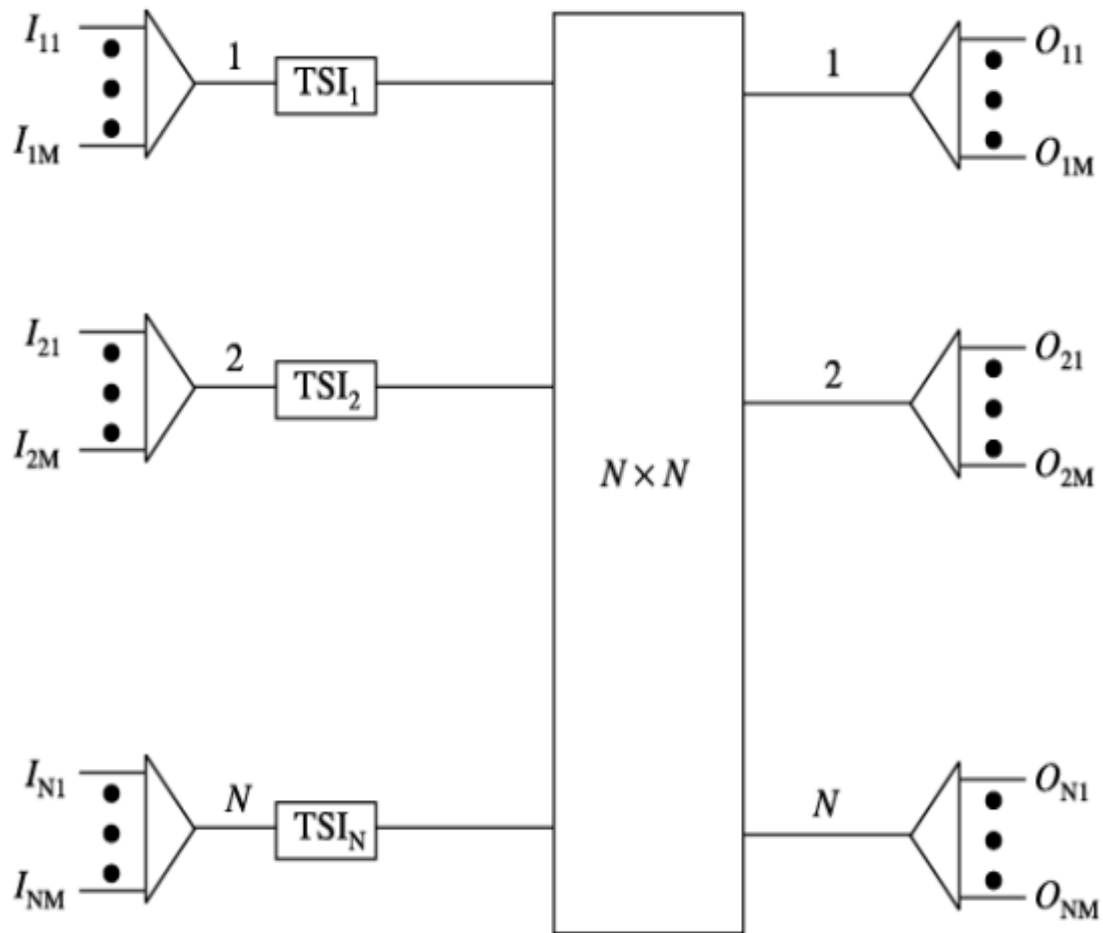
CM = control memory DTS = time slot clock DM = data memory

(a) Principle of time slot interchange

Combination switch

- Time slot interchange switches are not capable of switching sample values across the trunks without the help of some space switching matrices.
- a combination of the time and space switches leads to configurations that achieve both time slot interchange and sample switching across trunks.
- A two-stage combination switch may be organised with time switch as the first stage and the space switch as the second stage or vice versa.
- Accordingly the two switch configurations are known by the nomenclature time-space (TS) or space-time (ST) switches respectively.

Combination switch



Two-stage TS switch.

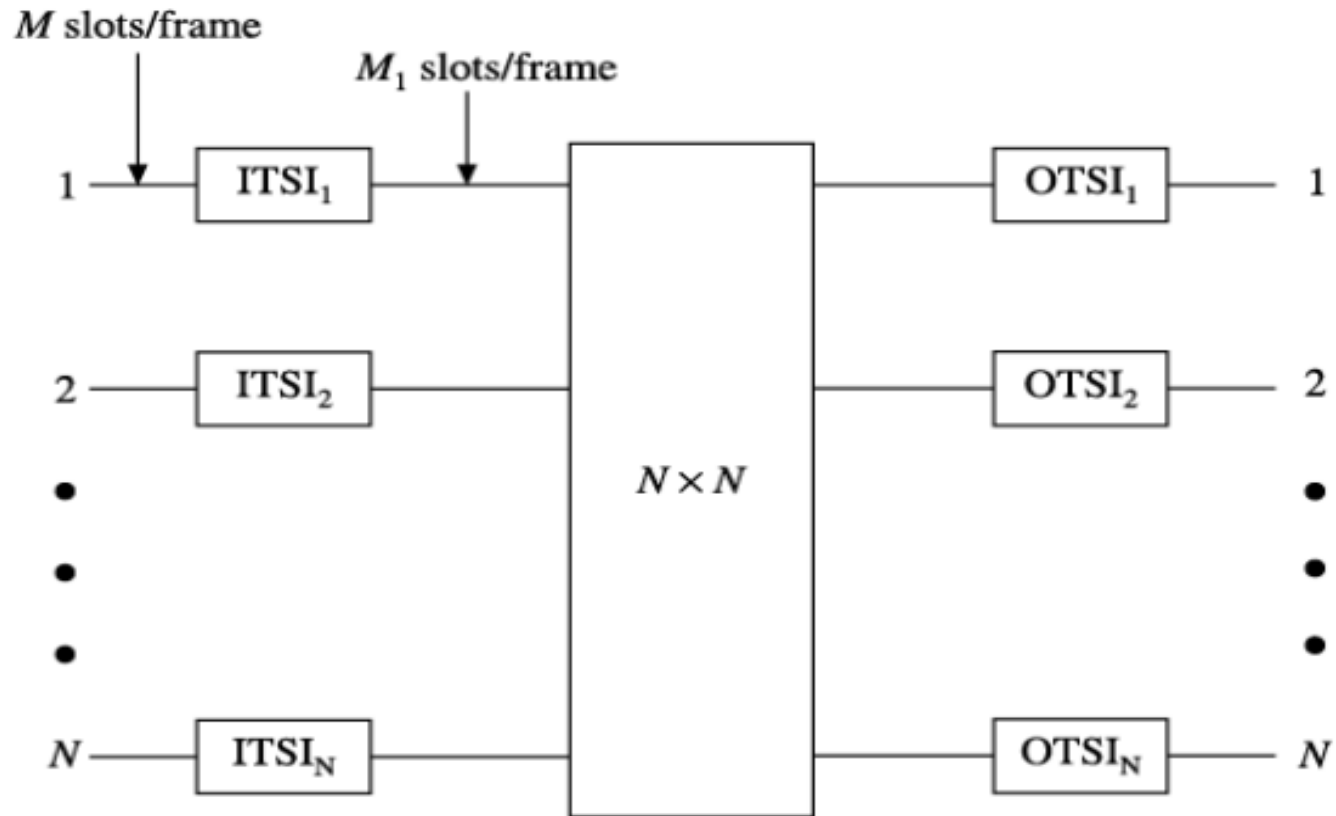
Three stage combination switching

Three-stage time and space combination switches are more flexible than their two-stage counterparts

The most common three-stage configurations are:

- Those which place time stages on either side of a space stage giving rise to TS configuration
- Those which place space stages on either side of a time stage giving rise to STS configuration

Three stage combination switching



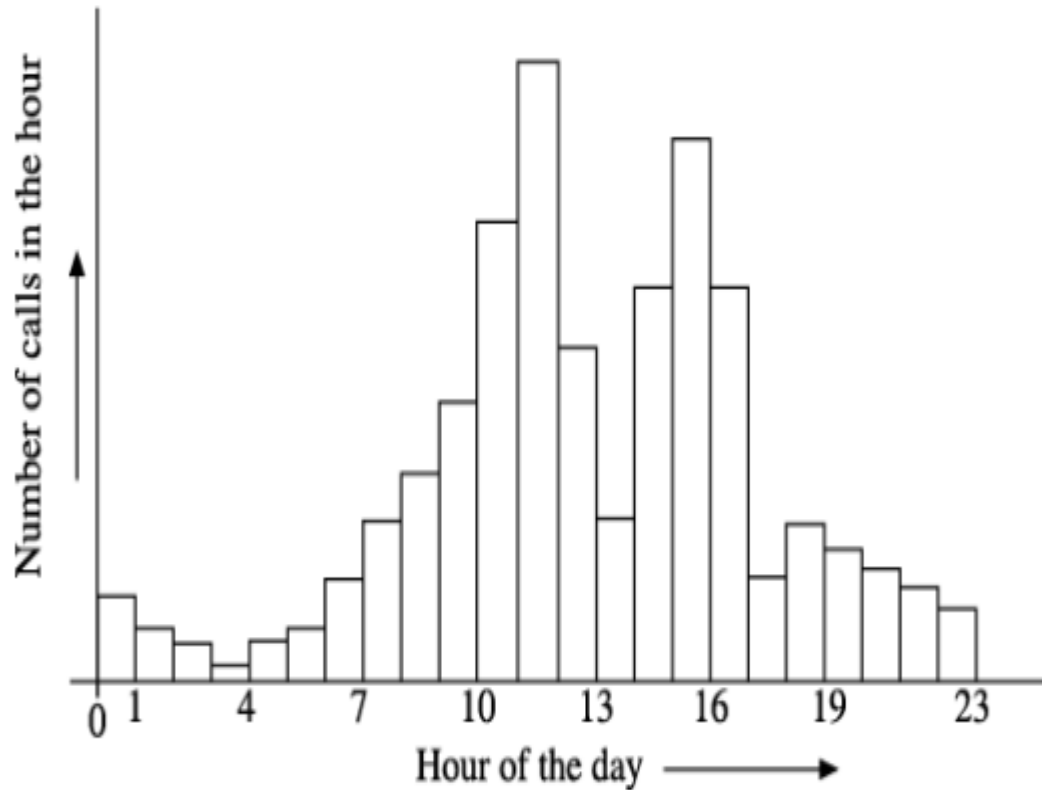
n-stage combination switching

- Very large time division switches, supporting 40,000 lines or more. Can be economically designed by using more than three stages of time and space combination switching.
- This N-stage switching where N is greater than three.
- N-stage combination networks may also be designed to provide better flexibility and less blockage.
- Basically two approaches are seen in designing N-stage switches
 - ✓ Expanding a TST switch with additional space stages
 - ✓ Modular design using a number of time-space (TS) modules.

Network traffic load and parameters

- There is a large peak around mid-forenoon and mid-afternoon signifying busy office activities.
- The afternoon peak is however slightly smaller.
- The load is low during the lunch-hour period. i.e. 12.00—14.00 hours.
- The period 17.00—18.00 hours is characterised by low traffic signifying that the people are on the move from offices to their residences.

Network traffic load and parameters



Typical telephone traffic pattern on a working day.

Network traffic load and parameters

- **Busy Hour:** Continuous 1-hour period laying wholly in the time interval concerned, for which the traffic volume or the number of call attempts is greatest.
- **Peak Busy Hour:** The busy hour each day: it usually varies from day to day. or over a number of days.
- **Time Consistent Busy Hour:** The 1-hour period starting at the same time each day for which the average traffic volume or the number of call attempts is greatest over the days under consideration.

Grade of service

- In loss systems. the traffic carried by the network is generally lower than the actual traffic offered to the network by the subscribers.
- The overload traffic is rejected and hence is not carried by the network.
- The amount of traffic rejected by the network is an index of the quality of the service offered by the network.
- This is termed grade of service (GOS) and is defined as the ratio of lost traffic to offered traffic Offered traffic is the product of the average number of calls generated by the users and the average holding time per call.

Grade of service

➤ GOS is given by

$$\text{GOS} = \frac{A - A_0}{A}$$

where

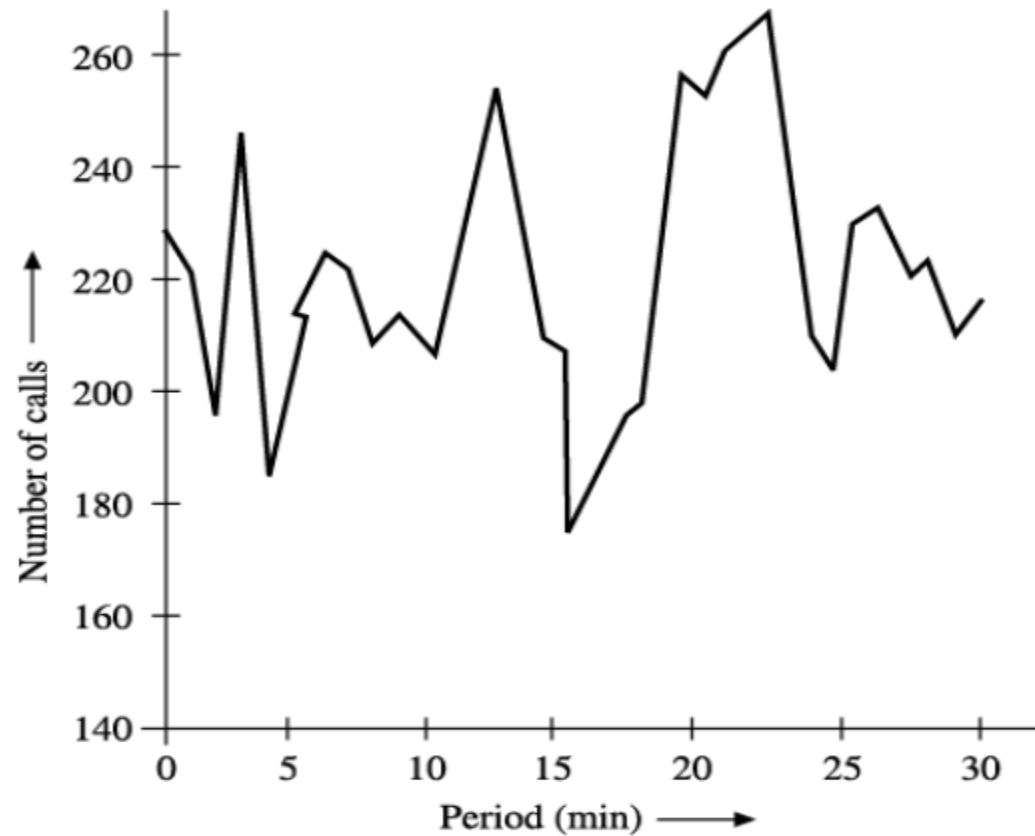
A = offered traffic

A_0 = carried traffic

$A - A_0$ = lost traffic

The smaller the value of grade of service, the better is the service. The recommended value for GOS in India is 0.002 which means that two calls in every 1000 calls or one call in every 500 calls may be lost. Every common subsystem in a network has an associated GOS value.

Modeling switching systems



Typical fluctuations in the number of telephone calls.

Modeling switching systems

- A telecommunication network carries traffic generated by a large number of individual subscribers connected to the network.
- The call generation by the subscribers and therefore the behaviour of the network or the switching systems in it can be described as a random process.

Modeling switching systems

- Subscribers generate calls in a random manner.
- A random process or a stochastic process is one in which one or more quantities vary with time in such a way that the instantaneous values of the quantities are not determinable precisely but are predictable with certain probability.

Modeling switching systems

Similarly, the time index of the random variables can be discrete or continuous.

Accordingly, we have four different types of stochastic processes:

1. Continuous time continuous state
2. Continuous time discrete state
3. Discrete time continuous state
4. Discrete time discrete state.

Incoming traffic and service time characterization

- Whenever a subscriber originates a call, he adds one to the number of calls arriving at the network and has no way by which he can reduce the number of calls that have already arrived.
- We are thus in need of a model that describes an originating process.
- Interestingly, this process can be treated as a special case of the B-D process in which the death rate is equal to zero.
- In other words, there is no death occurring in the process.
- Such a process is known as a renewal process.

Blocking models and loss estimates

- The behaviour of loss systems is studied by using blocking models and that of the delay systems by using queuing models.
- it is apparent that we are concerned with three aspects while dealing with the analysis of the telecommunication systems:
 1. Modelling the system
 2. Traffic arrival model
 3. Service time distribution

Blocking models and loss estimates

- In loss systems. the overflow traffic is rejected.
- In other words the overflow traffic experiences blocking from the network.
- There are three ways in which overflow traffic may be handled
 - ✓ The traffic rejected by one set of resources may be cleared by another set of resources in the network.
 - ✓ The traffic may return to the same resource after sometime.

Blocking models and loss estimates

Some part of the initial information from the subscriber may be lost. But the resources are allocated soon enough so that the loss is unnoticeable by the subscriber or within acceptable limits.

Corresponding to the above three cases, we consider three models of loss systems:

1. Lost calls cleared (LCC)
2. Lost calls returned (LCR)
3. Lost calls held (LCH)

Delay systems

- A class of telecommunication networks. such as data networks, places the call or message arrivals in a queue in the absence of resources, and services them as and when resources become available.
- Servicing is not taken tip until the resource becomes available.
- Such systems are known as delay systems which are also called **lost call delayed** (LCD) systems.

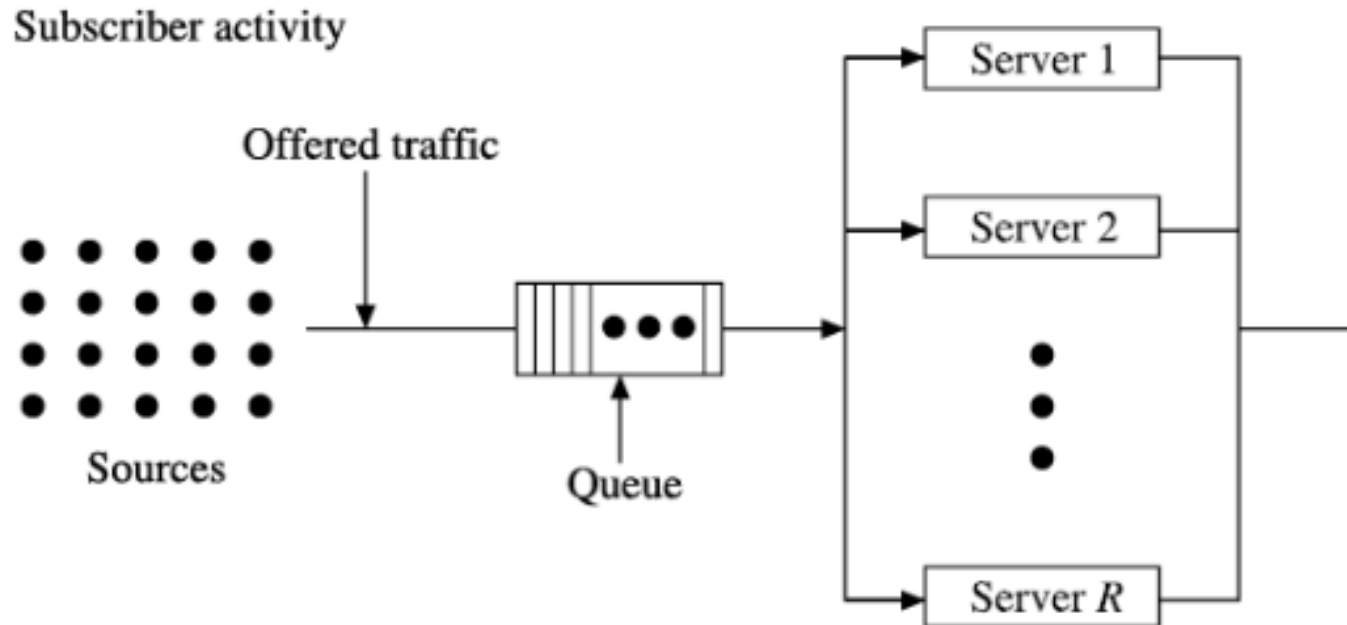
Delay systems

- Delay systems are analysed using queuing theory which is sometimes known as waiting line theory.
- Although the foundations of queuing theory were laid by early tele traffic researchers, the theory is nowadays used for the analysis of a wide variety of applications outside telecommunications.

Delay systems

- Examples of delay systems in telecommunications include the following:
 - . Message switching
 - . Packet switchin2
 - . Digit receiver access
 - . Automatic call distribution
 - . Call processing

Delay systems



Elements of a queuing system.